

Eye for AI

Supporting Critical AI Literacy in Small Business Food Marketing

Introduction

According to the U.S. Chamber of Commerce, as of 2025 58% of small businesses used generative AI, up from 40% in 2024 and 23% in 2023 (U.S. Chamber of Commerce, 2025). This growth is accommodated by outreach initiatives such Small Business B(AI)sics, an online training promoting basic competency and adoption (U.S. Chamber of Commerce, n.d.). As an introductory effort, it balances basic familiarity with considerations about privacy, copyright and awareness of AI’s limitations, but leaves small businesses who are getting started with AI ill-prepared to navigate the socially constructed perception around the use of AI, especially pertaining to public facing products and materials.

The purpose of this campaign supplement, therefore, is for business owners to appreciate the complexity of the reception their marketing containing AI outputs may provoke by developing a sensitivity of the public’s “Eye for AI” in the democratized arena of judging “good food” (Feldman, 2021). In fact, by raising their awareness of technology’s social dimension at its still nascent stage, small businesses may find that their own AI taste buds become exercised to the point of judging generated outputs in a more nuanced and holistic way.

AI food image retrospective

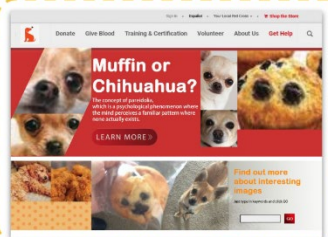
Inputs		Outputs of GPT-4V	Interest in the neural network architecture underlying today's generative AI received a significant boost from the 2012 ImageNet Large Scale Visual Recognition Challenge where AlexNet, the sole neural network entrant to the competition, achieved a new state-of-the-art error rate of 15.3%, substantially outperforming the second-place system's 26.2% error rate (Krizhevsky et al., 2012). Since then, the "Chihuahua or muffin?" comparison task, first introduced and popularized on social media (Zack, n.d.), has become a familiar illustration of the ambiguities involved in machine vision with Fan et al. (2024) recently demonstrating that even today's advanced models may fail image classification within a multi-panel layout (Figure 1).
Single-panel image	 Does this image show a muffin?	Yes 	
Multipanel image (Poster)	 Does the middle subfigure in the bottom row show a muffin?	No 	
Multipanel image (Web Screenshot)	 Does the top right image show a muffin?	No 	
 Answer correct		 Answer wrong	

Figure 1: Illustration of the ongoing challenges, as of 2024, regarding object recognition in complex layouts (from Fan et al., 2024)

This compound classification failure underscores the strangeness of the intelligence exhibited by AI systems which, highlighting their conspicuous limitations and widely publicized mistakes, excel at the completion of a prediction task only up to a certain limit of constraints. Their performance can therefore appear simultaneously impressive and strangely incongruent depending on the number of competing requirements. For food marketing this means that an AI system will almost certainly complete the task of generating a recognizable image while failing to accommodate the layered expectations customers and the public at large have especially for locally owned and operated businesses in their neighborhood.

AI food image backlash

The need for a critical literacy supplement to the Chamber of Commerce's online campaign is motivated by an emerging reality for food businesses finding that AI-generated images can evoke unexpected reactions as evidenced in popular media coverage of backlash against image generation outputs in restaurant food displays (Yadegaran, 2026). Providing evidence for this phenomenon from the scientific literature, in a recent experimental study Diel et al. (2025, n=95) point out that AI outputs are rated as particularly uncanny as compared to clearly unrealistic or actual images. Hypothesizing food specific arousal of eeriness related to disgust or novelty, the researchers found this effect was predicted by the rater's propensity for food neophobia, a concept associated with unease about unfamiliar and uncertain outcomes.



Figure 2: San Francisco store front following controversy of AI generated food images (from Yangeran, 2026)

That businesses, especially small businesses, may be placing themselves on an axis of mistrust and even disgust as well as outside of the realm of health and naturalness is further illustrated by Stright et al. (2025, n=154) who found that “participants' perception of AI was negatively correlated to participants' willingness to consume the food product, as well as their perception of the healthiness, naturalness, and appeal of the product” (pp. 1). Therefore, when considering use of AI for marketing materials or other representative, directly customer-facing outreach, it is important to keep in mind the extent to which the generated image will mediate the relationship between business and the public.

AI in the neighborhood

With regard to the particular vulnerability of locally operated small businesses, Yang & Aggarwal (2019) argue empirically that consumers expect small companies to exhibit greater community orientation than large companies and may judge them more negatively when their behavior violates expectations of warmth, sincerity, and concern for others. AI and algorithmic operations more generally have shown to trigger these heightened demands, although not uniformly across domains but instead specifically concerning the performance of subjective tasks (Castelo et al., 2019) and hedonic services (Belanche et al., 2025). Delegating store front marketing to AI marks such a conspicuous crossing into the subjective where customers regard visual marketing as an expression of judgment, care, and personal involvement. This is further contextualized

by the perspectives shared in Puntoni et al. (2021) where a consumer situated in the “AI ecosystem” may experience a business owner’s task delegation as symbolic of replacement and alienation and may even be triggered to relive or associate concrete experiences of exploitation and misunderstanding (Figure 3).

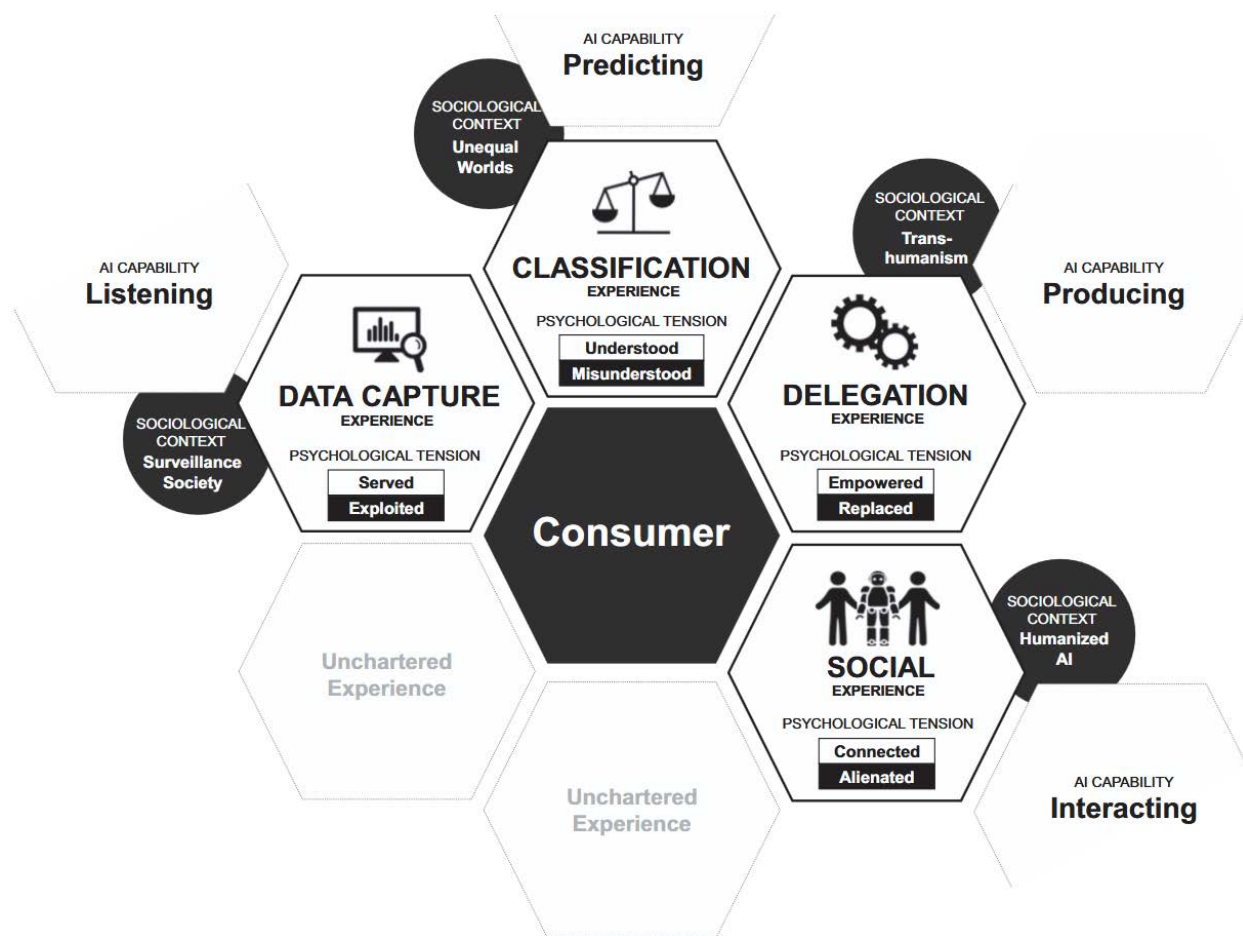


Figure 3: AI “ecosystem” capabilities and risks, (from Puntoni et al. 2021)

Conclusion

As businesses adopt AI to pursue growth, cost savings, and operational efficiency, those objectives may be undermined by insufficient attention to how AI use is perceived by customers and communities. Existing chamber-of-commerce outreach emphasizes practical adoption, streamlined work, efficiency, and business growth. Supplementing these promotional and capability-oriented initiatives with critical AI literacy would help business owners understand the social context of emerging technologies, anticipate potential points of friction, and avoid preventable public backlash. Each public-facing use of AI also contributes to the norms through which later applications will be interpreted. Rather than treating AI as a drop-in solution generator, businesses should approach it with an emphasis on co-creation and critical engagement. These practices allow owners to evaluate not only whether an output satisfies a short-term business need, but also whether it accurately represents the business’ identity and communicates the care, authenticity, and accountability expected by public.

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